

Debugging Closest Pair Code

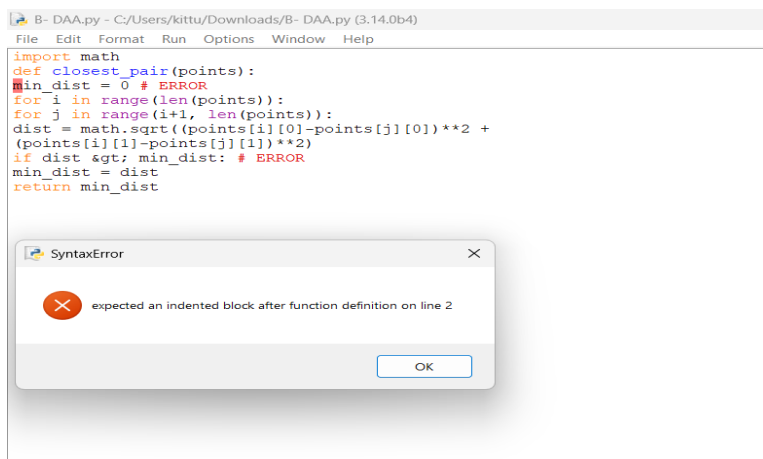
Q3. Debugging Closest Pair Code

The following Closest Pair brute force code has logical issues in distance calculation and comparison. Identify all errors and explain why incorrect results occur. Debug the code step-by-step to ensure accurate distance computation. Optimize the implementation by avoiding redundant calculations. Analyze the time complexity of the corrected version. Rewrite the final code with proper comments.

Original Program

```
import math

def closest_pair(points):
    min_dist = 0 # ERROR
    for i in range(len(points)):
        for j in range(i+1, len(points)):
            dist = math.sqrt((points[i][0]-points[j][0])**2 +
                              (points[i][1]-points[j][1])**2)
            if dist > min_dist: # ERROR
                min_dist = dist
    return min_dist
```



Error 1

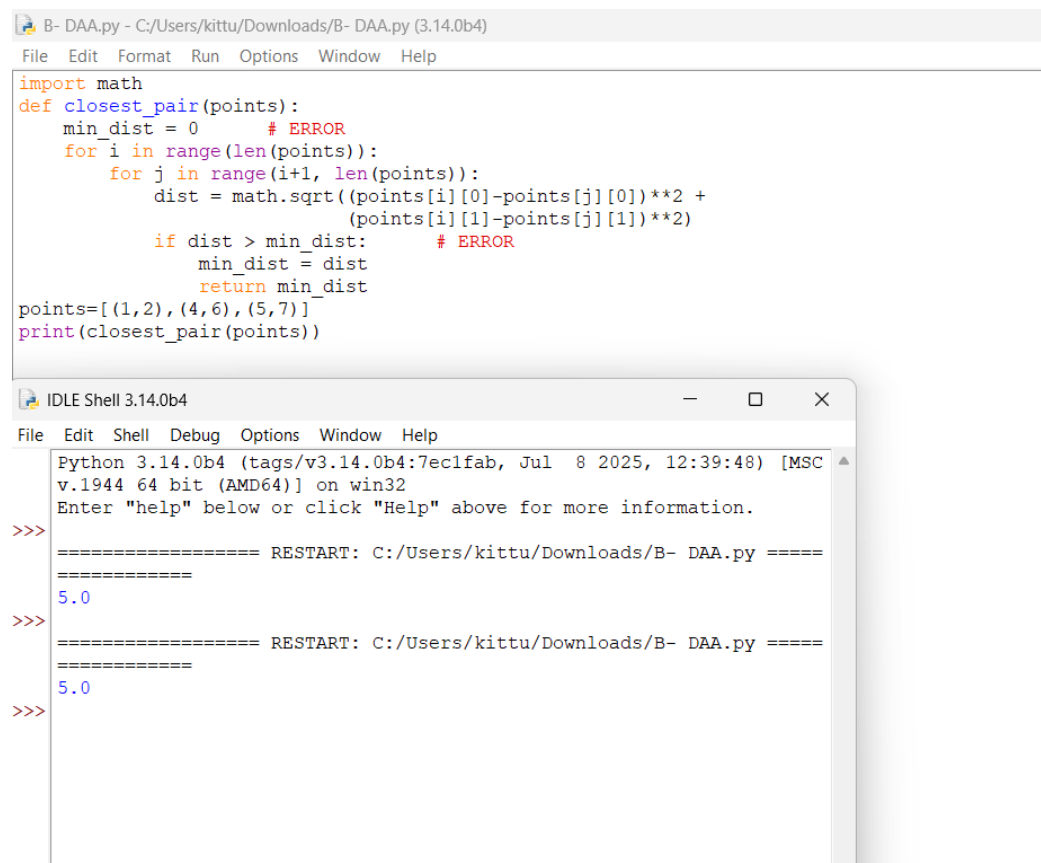
What is Wrong?

The code inside the function is **not indented**. Python expects all statements inside the function to have indentation.

Rectification

Indent all statements inside the function using **4 spaces (or one Tab)**.

After Correcting Indentation



The screenshot displays a Python IDE window titled 'B- DAA.py - C:/Users/kittu/Downloads/B- DAA.py (3.14.0b4)'. The code in the editor is as follows:

```
import math
def closest_pair(points):
    min_dist = 0 # ERROR
    for i in range(len(points)):
        for j in range(i+1, len(points)):
            dist = math.sqrt((points[i][0]-points[j][0])**2 +
                             (points[i][1]-points[j][1])**2)
            if dist > min_dist: # ERROR
                min_dist = dist
        return min_dist
points=[(1,2),(4,6),(5,7)]
print(closest_pair(points))
```

Below the editor is the 'IDLE Shell 3.14.0b4' window. It shows the Python version and system information, followed by two execution attempts. Both attempts result in a 'RESTART' message and output the value '5.0'.

```
Python 3.14.0b4 (tags/v3.14.0b4:7ec1fab, Jul 8 2025, 12:39:48) [MSC
v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/kittu/Downloads/B- DAA.py =====
5.0

>>>
===== RESTART: C:/Users/kittu/Downloads/B- DAA.py =====
5.0

>>>
```

Error 2

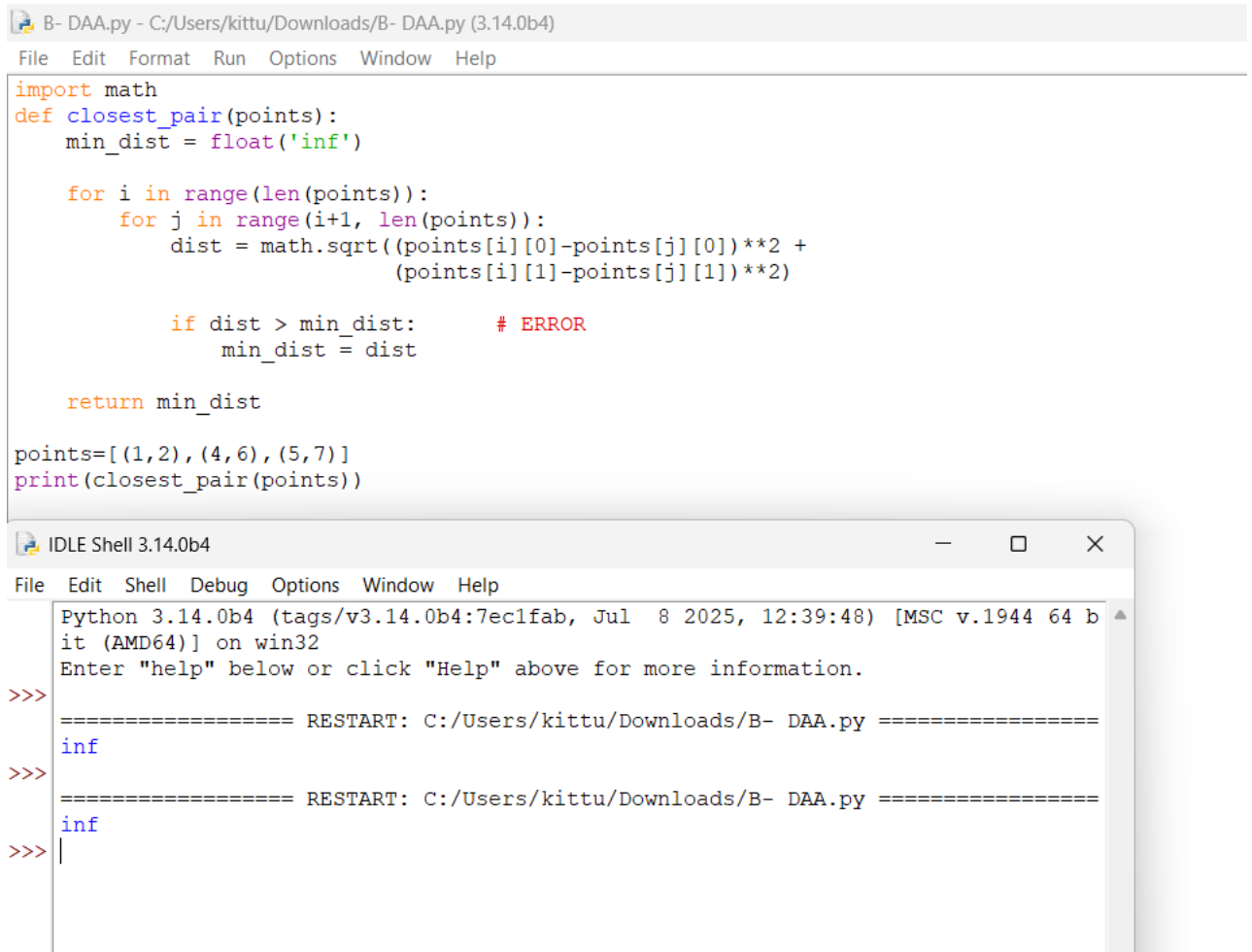
What is Wrong?

The `return min_dist` statement is inside the `for` loop. Therefore, the function stops after the first iteration and does not compare all pairs of points.

Rectification

Move `return min_dist` outside the `for` loop.

After Moving return



The screenshot shows a Python IDE window titled 'B- DAA.py - C:/Users/kittu/Downloads/B- DAA.py (3.14.0b4)'. The code defines a function `closest_pair` that calculates the minimum distance between points. It contains a logic error in the comparison operator. Below the code, the IDLE Shell window shows the execution output, which includes a restart message and the output `inf`.

```
import math
def closest_pair(points):
    min_dist = float('inf')

    for i in range(len(points)):
        for j in range(i+1, len(points)):
            dist = math.sqrt((points[i][0]-points[j][0])**2 +
                             (points[i][1]-points[j][1])**2)

            if dist > min_dist:      # ERROR
                min_dist = dist

    return min_dist

points=[(1,2), (4,6), (5,7)]
print(closest_pair(points))
```

```
Python 3.14.0b4 (tags/v3.14.0b4:7ec1fab, Jul  8 2025, 12:39:48) [MSC v.1944 64 b
it (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kittu/Downloads/B- DAA.py =====
inf
>>>
===== RESTART: C:/Users/kittu/Downloads/B- DAA.py =====
inf
>>> |
```

Error 3

What is Wrong?

The comparison operator is incorrect.

The program checks for the greater distance instead of the smaller distance.

Rectification

Change

`if dist > min_dist:` to `if dist < min_dist:`

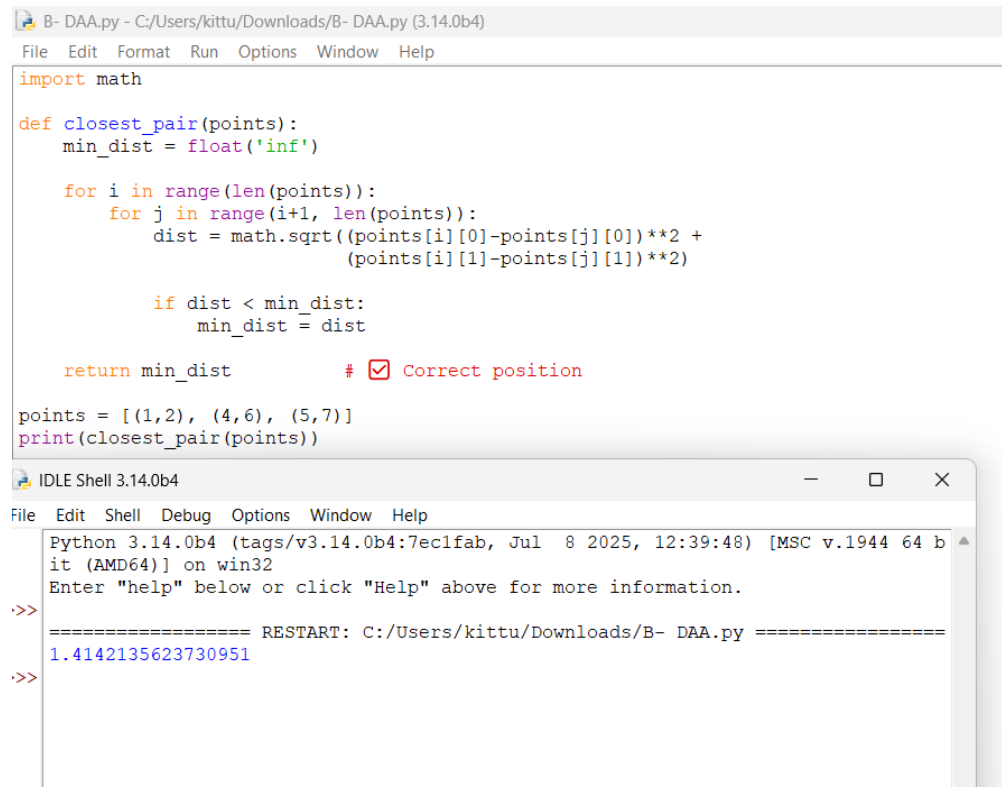
FINAL PROGRAM

```
import math

def closest_pair(points):
    min_dist = float('inf')
    for i in range(len(points)):
        for j in range(i + 1, len(points)):
            dist = math.sqrt((points[i][0] - points[j][0])**2 +
                              (points[i][1] - points[j][1])**2)
            if dist < min_dist:
                min_dist = dist
    return min_dist

points = [(1,2), (4,6), (5,7)]

print(closest_pair(points))
```



The screenshot shows a Python IDE window titled "B- DAA.py - C:/Users/kittu/Downloads/B- DAA.py (3.14.0b4)". The code editor contains the same Python code as shown in the previous block. Below the code editor is the "IDLE Shell 3.14.0b4" window, which displays the output of the program. The shell shows the Python version and architecture, followed by the execution of the code. The output is "1.4142135623730951", which is the minimum distance between the points (1,2), (4,6), and (5,7).

```
B- DAA.py - C:/Users/kittu/Downloads/B- DAA.py (3.14.0b4)
File Edit Format Run Options Window Help
import math

def closest_pair(points):
    min_dist = float('inf')
    for i in range(len(points)):
        for j in range(i+1, len(points)):
            dist = math.sqrt((points[i][0]-points[j][0])**2 +
                              (points[i][1]-points[j][1])**2)
            if dist < min_dist:
                min_dist = dist
    return min_dist      # ☒ Correct position

points = [(1,2), (4,6), (5,7)]
print(closest_pair(points))

IDLE Shell 3.14.0b4
File Edit Shell Debug Options Window Help
Python 3.14.0b4 (tags/v3.14.0b4:7ec1fab, Jul  8 2025, 12:39:48) [MSC v.1944 64 b
it (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kittu/Downloads/B- DAA.py =====
1.4142135623730951
>>>
```

Final Explanation

Corrections Made

- Corrected indentation inside the function.
- Changed `min_dist = 0` to `min_dist = float('inf')`.
- Changed `if dist > min_dist` to `if dist < min_dist`.

Algorithm: Closest Pair (Brute Force)

Input: List of points (x, y)

Output: Minimum (closest) distance between any two points.

Steps:

1. Start.
2. Initialize `min_dist = float('inf')`.
3. For each point `i`, compare it with every point `j` where `j = i + 1`.
4. Calculate the Euclidean distance:

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

5. If the calculated distance is less than `min_dist`, update `min_dist`.
6. Repeat until all point pairs are compared.
7. Return `min_dist`.
8. Stop.

Example

Input:

`points = [(1,2), (4,6), (5,7)]`

Distance Calculations

- Distance between (1,2) and (4,6)

$$\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5.0$$

- Distance between (1,2) and (5,7)

$$\sqrt{(5-1)^2 + (7-2)^2} = \sqrt{16+25} = \sqrt{41} = 6.4031$$

- Distance between (4,6) and (5,7)

$$\sqrt{(5-4)^2 + (7-6)^2} = \sqrt{1+1} = 1.4142$$

The **closest pair** is (4,6) and (5,7).

Minimum Distance = 1.4142

Optimization Technique

1. Compare each pair of points only once by using $j = i + 1$.
2. Initialize `min_dist` to infinity (`float('inf')`) so the first calculated distance becomes the initial minimum.
3. Update `min_dist` only when a smaller distance is found.
4. Return the result after all point pairs are compared.
5. This avoids redundant comparisons and ensures the correct closest distance is found.

Time Complexity

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$

Result

The corrected program successfully finds the minimum (closest) distance between all pairs of points.